Transfer: momentum and impulse

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Apply momentum reasoning to new collisions and deflection cases.

SUCCESS CRITERIA - I can...

1. Define the system and positive direction.
2. Use conservation or impulse only when the facts needed for the equation are true.
3. Explain the energy changes and the limits of your calculation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

In the same calculation for straight-line deflection, keep sideways Δv fixed but make lead time **3** times as long. **Compare shift** and **name a limitation**.

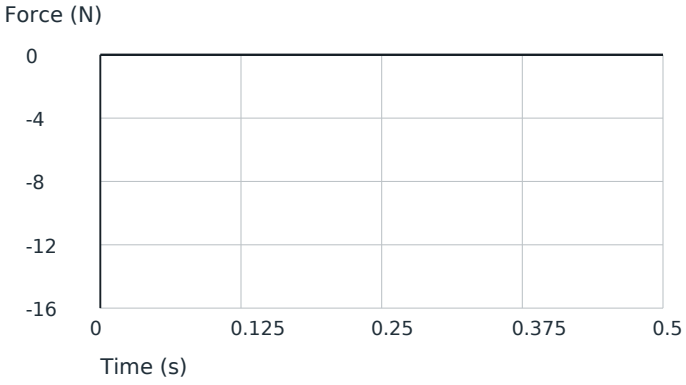
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p = mv$; $J = \Delta p = F_{avg}\,\Delta t$; $p_{total,\,i} = p_{total,\,f}$.
Use momentum conservation only when external impulse is negligible. **Label before**/after diagrams and directions.

Worked reasoning: A **2 kg** cart moving at **3 m/s** stops in **0.5 s**. Momentum change is **-6 kg·m/s**; average force is **-12 N**. Force direction opposes the initial motion.

Your turn: Represent the same stop with momentum arrows and a force-time rectangle.

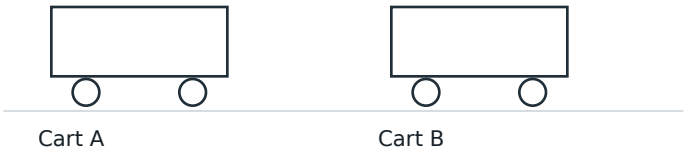


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A **1 kg** cart at **4 m/s** right sticks to a **1 kg** cart at rest. **Draw before**/after and **explain which quantity** is conserved.

Before contact: add velocity arrows.



Marzano check - circle: ① ② ③

04 Apply the physics

A **2 kg** cart moving right at **6 m/s** stops in **5 s**.
1. **Calculate its momentum-change magnitude** and average stopping-force magnitude.
2. Use the before-and-after diagram to **show the direction of the** change.

Before	**After**

Label the initial and final momentum.

Marzano check - circle: ① ② ③

Transfer: momentum and impulse

antocciphysics

Continue down each column.

05 Explain your reasoning

A report says a collision is elastic because total momentum is equal before and after. What extra calculation is needed to **check this conclusion**?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 kg** cart travels left at **6 m/s**. **Find momentum** and **draw its direction**.
Use before and after snapshots to organize the momentum account.

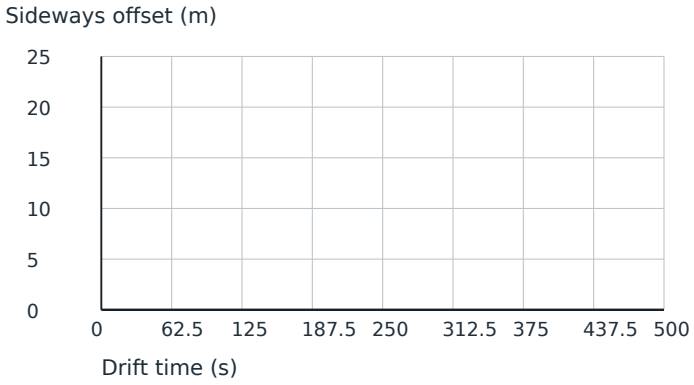
Before	**After**

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

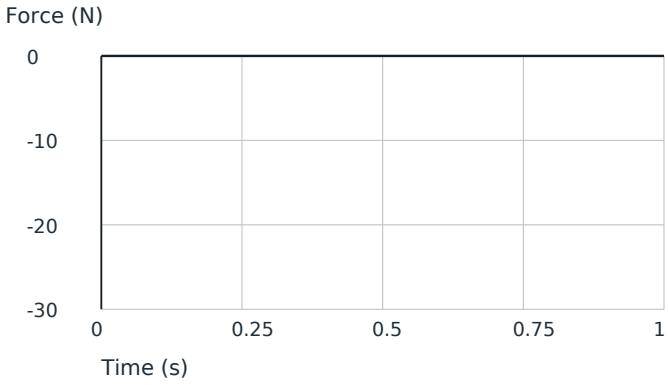
A fixed impulse gives a target **0.04 m/s** sideways velocity change. **Find shift after 500 s** and **state a part of the** explanation that may not fit **what happens**.



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Return to the worked example in problem **2**.
1. Design two stopping processes for this cart that give the same impulse but different average forces.
2. **Explain the time-force tradeoff**.



Marzano check - circle: 1 2 3